

Jiachuan Wang

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SUMMARY:

My research focuses on computational and cognitive neuroscience, with an emphasis on understanding the neural mechanisms of memory and decision-making. I am currently pursuing a Ph.D. at the National University of Singapore, where I develop biologically plausible models of associative memory and navigation. My previous work combined neuroimaging, machine learning, and computational modeling to investigate human cognition and large-scale brain organization.

EDUCATION:

National University of Singapore <i>Ph.D. candidate in Medicine</i> <i>Neuroscience Track and Biostatistics, Bioinformatics & Epidemiology Track</i>	Singapore Aug. 2023 – Present
Zhejiang University (Joint Degree Programme) <i>B.S. in Bioinformatics</i>	Hangzhou, China Sep. 2019 – Jun. 2023
The University of Edinburgh <i>B.S. (Hons) in Biomedical Informatics</i>	Edinburgh, UK Sep. 2019 – May. 2023

EXPERIENCE:

The N.1 Institute for Health, National University of Singapore <i>Graduate Researcher</i> <i>Advisors: Andrew Tan, Camilo Libedinsky, Shih-Cheng Yen</i>	Singapore Sep. 2023 – Present
Centre for Discovery Brain Sciences, The University of Edinburgh <i>Research Intern; Advisor: Gediminas Lukšys</i>	Edinburgh, UK Mar. 2022 – Aug. 2023
School of Brain Science and Brain Medicine, Zhejiang University <i>Research Intern; Advisor: Zhiping Wang</i>	Hangzhou, China Jan. 2022 – Sep. 2022 Jun. 2020 – Aug. 2020
Interdisciplinary Institute of Neuroscience and Technology, Zhejiang University <i>Research Intern; Advisor: Anna Wang Roe</i>	Hangzhou, China Apr. 2021 – Oct. 2021

PUBLICATIONS:

- Kong, X., Shu, X., **Wang, J.**, Liu, D., Ni, Y., Zhao, W., Wang, L., Gao, Z., Chen, J., Yang, B., Guo, X. and Wang, Z. (2022) Fine-tuning of mTOR signaling by the UBE4B-KLHL22 E3 ubiquitin ligase cascade in brain development. *Development*. doi: 10.1242/dev.201286.
- Zhang, L., Ma, X., Wu, Z., Liu, J., Gu, C., Zhu, Z., **Wang, J.**, Shu, W., Li, K., Hu, J. and Lv, X. (2022) Prevalence of ground glass nodules in preschool children: a cross-sectional study. *Translational Pediatrics*. doi: 10.21037/tp-22-465.

SELECTED CONFERENCE PRESENTATIONS:

- **Wang, J.**, Shetru, V. J., Kumar, M. G., Libedinsky, C., Yen, S.-C., Tan, A. Y.-Y. and Polepalli, J. S. Unsupervised Hebbian learning drives biologically interpretable pattern separation in a hippocampal–striatal network. Spotlight talk and poster presentations, **AAAI Workshop** on Neuro for AI & AI for Neuro, Singapore, January 2026. doi: 10.48448/jsed-nf31.
- **Wang, J.**, Shetru, V. J., Kumar, M. G., Libedinsky, C., Yen, S.-C., Tan, A. Y.-Y. and Polepalli, J. S. A biologically plausible computational model of hippocampal neurogenesis and pattern separation in memory. Poster presentation with short paper, 8th Annual Conference on **Cognitive Computational Neuroscience (CCN)**, Amsterdam, Netherlands, August 2025.
- Wang, S., **Wang, J.**, Zhu, W., Cheng, Y., Aydemir, B., Qiu, Y., Gerstner, W., Sandi, C. and Luksys, G. Computational model-based analysis of spatial navigation strategies under stress and uncertainty using place, distance and border cells. Poster presentation, **Society for Neuroscience (SfN)** Meeting, Washington, D.C., November 2023.

GRANTS & AWARDS:

NUS Research Scholarship <i>NUS Graduate School</i>	2023 – 2028
Outstanding Graduate of Zhejiang University <i>Zhejiang University</i>	2023
Zhejiang University Scholarship <i>Zhejiang University</i>	2022
Academic Scholarship (¥40,000) <i>ZJU-UoE Institute</i>	2022
Student Research Training Program Funding (¥1,200) <i>Zhejiang University</i>	2021 – 2022

TEACHING:

National University of Singapore <i>Graduate Teaching Assistant</i>	
• Systems Neurobiology	Fall 2024, Fall 2025
• Beginning Artificial Intelligence Through Neuroscience	Fall 2024

REVIEWING:

- Journal: *Scientific Reports*.
- Conference: Computational and Systems Neuroscience (COSYNE), Cognitive Computational Neuroscience (CCN).

ASSOCIATIONS:

- Society for Neuroscience, Singapore Chapter • ALBA Network